IRRIGATION WATER MANAGEMENT · NOTE SP02

Measuring Soil Moisture — Gravimetric Method, Bulk Density and Moisture Instruments

Course topics: IDE5.5 L10, IDE-515 T1

When and where to measure

- Measure or estimate soil moisture at regular intervals and at several depths in the root zone to know the rate of use and when and how much to irrigate. For most crops, start irrigating when about 50% of the available moisture in the root zone is depleted.
- Measure where the roots extract water, following the crop's extraction pattern: one reading in the upper quarter of the root zone and one or two lower down.

> Maximum extraction depth 60 cm → measure at 15 cm, 30 cm and 45 cm. While roots are young, 15 cm is usually enough; at maturity read all three depths.

Gravimetric (soil-sampling) method — the standard

1. Take soil samples from the desired depths at several locations.
2. Weigh them moist (Wm).
3. Oven-dry them at 105°C (105–110 °C for 24 hours) to constant weight and weigh again (Wd).
4. Express moisture as a percentage by weight on a dry basis.

$$\text{Moisture content (\%)} = (\text{Wm} - \text{Wd}) \,/\, \text{Wd} \times 100$$

Wm - Wd = weight of water; so moisture content = weight of water per unit weight of dry soil × 100.

> Moist sample 110 g, oven-dry 94 g: (110 - 94) / 94 × 100 = 17.02%.

Apparent specific gravity (bulk density)

- The ratio of the weight of a volume of dry soil (including pores) to the weight of an equal volume of water — also called volume weight or bulk density. Structure, texture and compactness affect it.
- Determine it on undisturbed samples of known volume taken with a core sampler, trimmed, oven-dried and weighed. Typical agricultural soils: 1.1 – 1.6 g cm^{-3}.

$$\text{Bulk density } (\text{g/cm}^3) = \text{weight of dry soil (g)} \,/\, \text{volume of soil } (\text{cm}^3)$$

$$\text{Moisture (cm per m depth)} = \text{apparent specific gravity} \times \text{moisture (\%)}$$

> At 17.02% moisture and bulk density 1.4: 1.4 × 17.02 ≈ 23.8 cm of water per metre of soil.

Instruments (indirect methods)

Commercial instruments are less accurate than sampling but good enough in the field.

Instrument	Principle and features	Limits
Tensiometer	A water-filled tube with a porous clay cup and a manometer or vacuum gauge. Installed at the desired depth, water moves out through the cup until it balances the soil; drier soil means higher tension and a higher reading. Reads in cm of water or hundredths of an atmosphere	Stops working above 0.85 atmosphere (air enters); good in sandy and sandy-loam soils, not in clay or cracking soils
Electrical-resistance blocks	Porous blocks (plaster of Paris, gypsum, nylon, fibreglass) with two embedded electrodes connected to a resistance or conductance meter; the block reaches moisture equilibrium with the soil and its resistance indicates moisture	Affected by soil salts and fertiliser; calibrate for each soil
Neutron moisture meter	Fast neutrons from a probe source are slowed by colliding with hydrogen nuclei; the count of slow neutrons at the detector, read on a portable battery scaler or ratemeter, is a direct function of the water present	Rapid, in-situ; monitors moisture through the whole root zone

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) lessons on soil–water–plant relations, evapotranspiration, irrigation efficiencies, and surface, sprinkler and drip irrigation. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Irrigation of most crops should begin when about ___ of available moisture is depleted

(a) 10%
(b) 25%
(c) 50%
(d) 90%

Q2. Soil samples for gravimetric moisture are dried at

(a) 60°C
(b) 80°C
(c) 105°C
(d) 150°C

Q3. A moist sample weighs 110 g and 94 g oven-dry; moisture content is

(a) 14.5%
(b) 16.0%
(c) 17.02%
(d) 21.3%

Q4. A tensiometer stops working when tension exceeds about

(a) 0.1 atm
(b) 0.85 atm
(c) 5 atm
(d) 15 atm

Q5. The neutron moisture meter works on

(a) electrical resistance
(b) neutron scattering by hydrogen
(c) capillarity
(d) osmosis

Answers

Q1 — (c) 50%
Q2 — (c) 105°C
Q3 — (c) 17.02%
Q4 — (b) 0.85 atm
Q5 — (b) neutron scattering by hydrogen